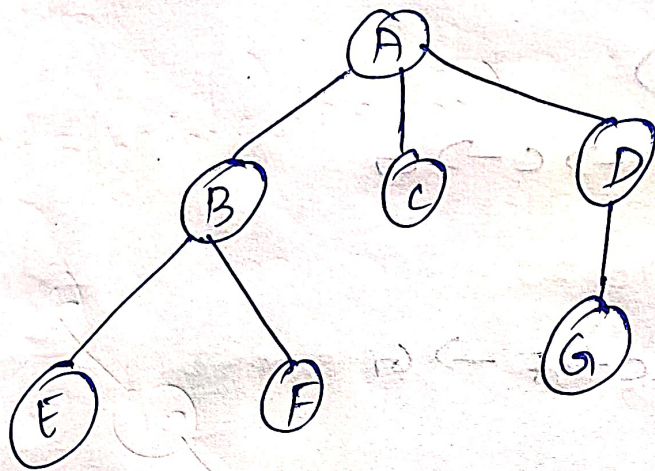


classroom Activity - Unit III

Trees

Activity I : Tree Terminology.

* Draw the tree: A as root, B, C, D as children, E, F under B, G under D.



* Identify: root, parent of G, children of B.

⇒ (i) root = A

(ii) parent of G = D

(iii) children of B = E, F

* Find: leaf nodes, Internal nodes, Degree of B, Height, level of F.

⇒ (i) leaf node = C, E, F, G

(ii) Internal node = A, B, D

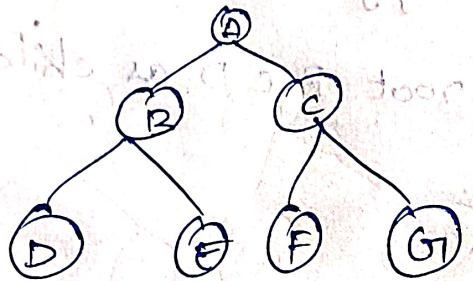
(iii) Degree of B = 2

(iv) Height = 2

(v) Level of F = 2

Activity 2: Traversal Race

- * Use Tree: A root; B, C children; D, E under B, F, G under C.



- * Write Inorder (LNR)

$D \rightarrow B \rightarrow E \rightarrow A \rightarrow F \rightarrow C \rightarrow G$

- * Preorder (NLR)

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

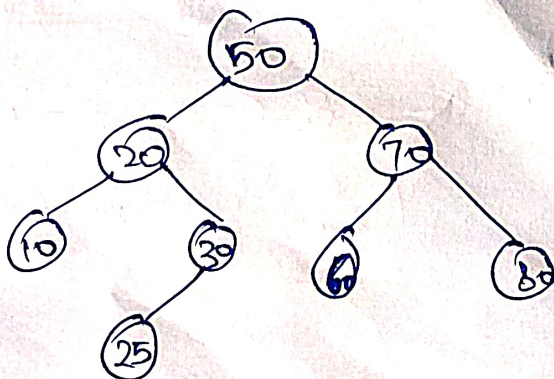
- * Postorder (LRN)

$D \rightarrow E \rightarrow B \rightarrow F \rightarrow G \rightarrow C \rightarrow A$

Activity-3: BST

- * Insert: 50, 20, 70, 10, 30, 60, 80, 25.

- * Draw the BST



* Find Minimum and Maximum

(i) minimum = 10

(ii) maximum = 80

* find Height and leaf nodes

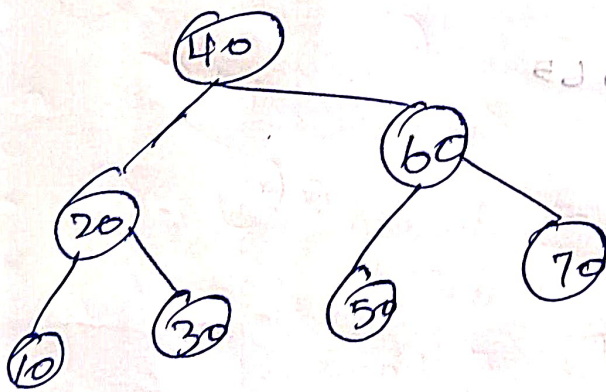
(i) Height = 3

(ii) leaf nodes = 10, 25, 60, 80



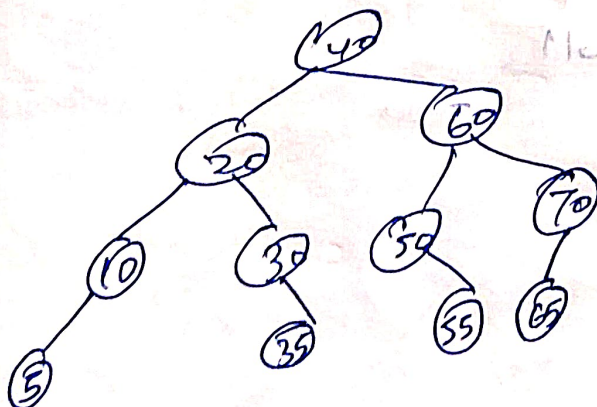
Activity 4: BST Insertion

* Initial BST: 40, 20, 60, 10, 30, 50, 70



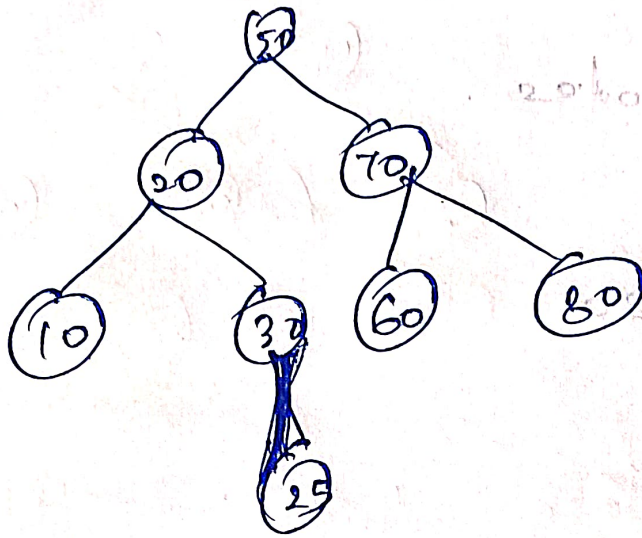
* Insert: 5, 55, 65, 55, 35

* Draw the updated BST



Activity 5: BST Search

Using activity 3 BST



* Search Value : 65, 90, 25

* For each search, write the nodes visited from the root

* Example : $50 \rightarrow 70 \rightarrow 60 \rightarrow 65$

(i) Search 65

$50 \rightarrow 70 \rightarrow 60 \rightarrow \text{Null}$

65 is not present in this BST.

(ii) Search 90

$50 \rightarrow 70 \rightarrow 80 \rightarrow \text{Null}$

90 is not found

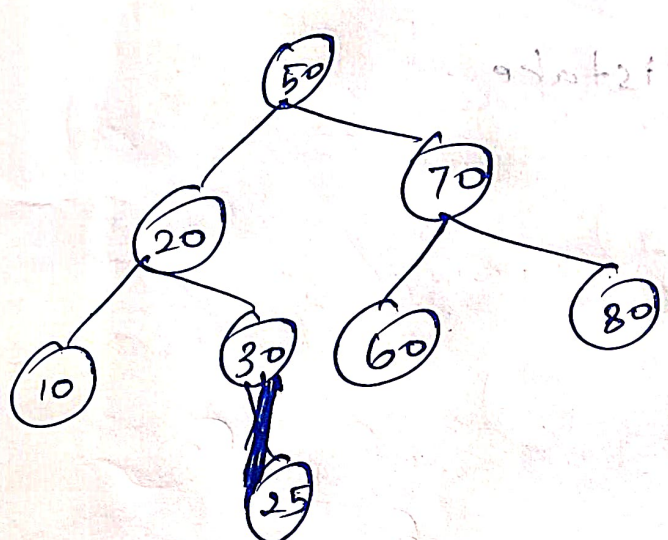
(iii) Search 25

50 → 20 → 30 → 25

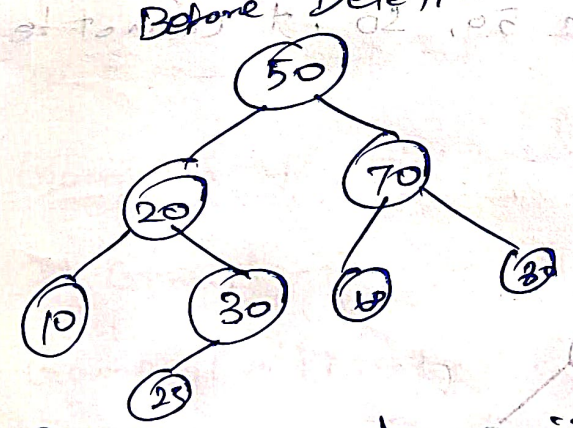
∴ 25 is found

Activity 6: BST Deletion.

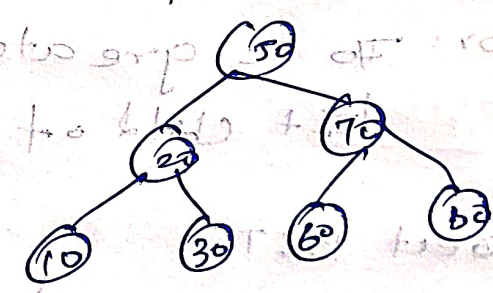
* Draw the BST (using Activity 3)



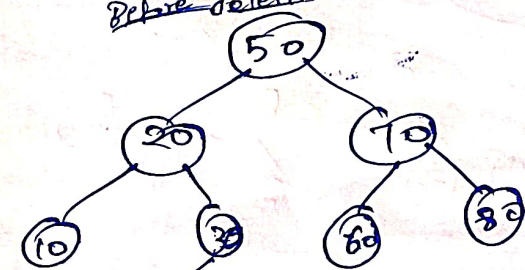
* Delete a leaf node (25)
Before Deletion



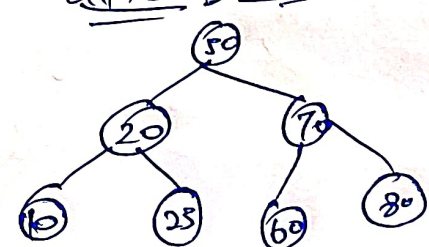
after Deletion



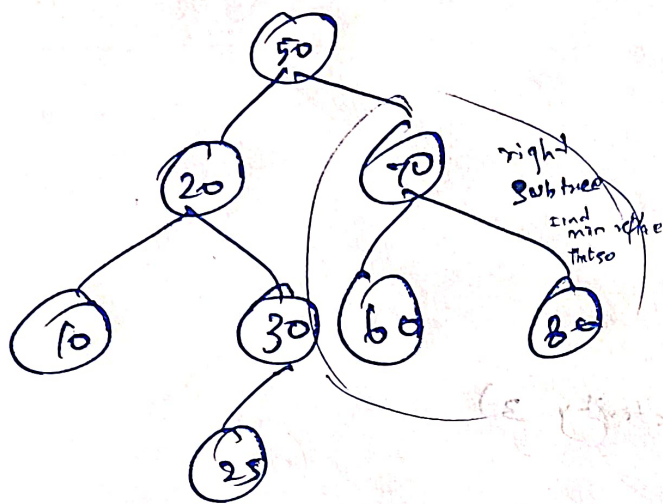
* Delete a node with one child (30)
Before deletion



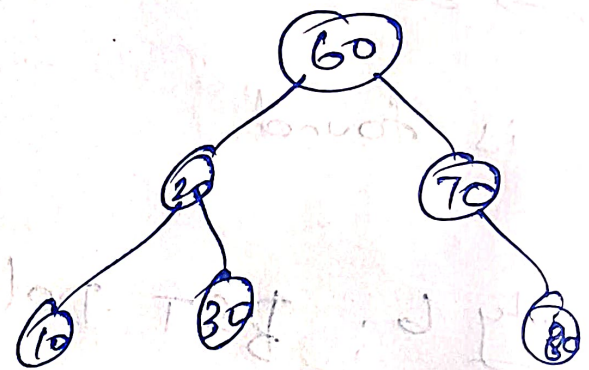
after Deletion



* Delete a Node w
Delete 50 (Before Deletion)

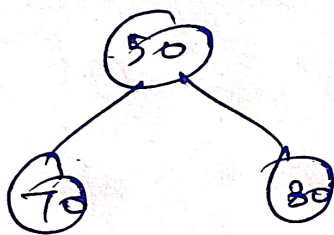


After Deletion.



Activity: 7: Find the Mistake.

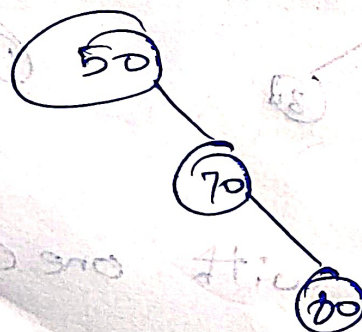
* Given incorrect BST:



* Identify the error and draw the correct

error: 70 is greater than 50, so it cannot be left child of 50

Correct BST:



Activity 8: Quad Tree

- Divide a 100×100 square into NW, NE, SE, SW.
- place point: $(10, 20)$, $(80, 70)$, $(30, 60)$, $(90, 10)$.
- Identify the correct quadrant for each point.

⇒

point	Quadrant
$(10, 20)$	SW
$(80, 70)$	NE
$(30, 60)$	NW
$(90, 10)$	SE

Activity 9: Application.

- File System $\rightarrow$ Tree
- Dictionary $\rightarrow$ BST
- Google Map $\rightarrow$ Quad Tree
- 3D Graph $\rightarrow$ Octree
- Game Collision $\rightarrow$ Octree

⇒

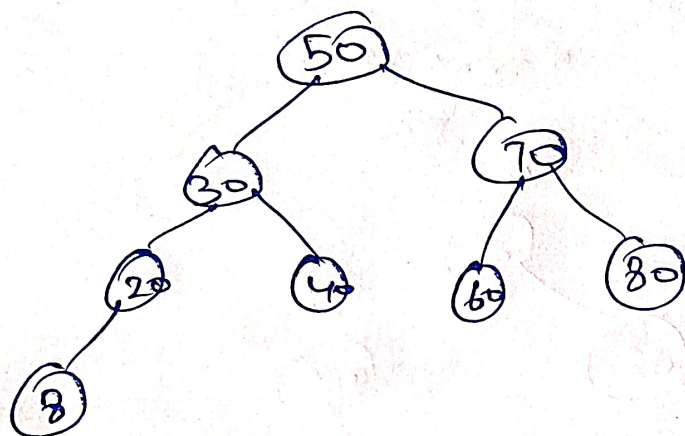
Application	Tree
File System	Tree
Dictionary	BST
Google map	Quad tree
3D Graph	Octree
Game Collision	Octree

Activity 10 : Mini Hackathon

* Construct a BST using Your assigned numbers.

⇒ Numbers : 50, 30, 70, 20, 40, 60, 8, 80

BTS:



* Perform Inorder, Preorder and Postorder

(i) Inorder

$8 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50 \rightarrow 60 \rightarrow 70 \rightarrow 80$

(ii) Preorder

$50 \rightarrow 30 \rightarrow 20 \rightarrow 8 \rightarrow 40 \rightarrow 70 \rightarrow 60 \rightarrow 80$

(iii) Postorder

$8 \rightarrow 20 \rightarrow 40 \rightarrow 30 \rightarrow 60 \rightarrow 80 \rightarrow 70 \rightarrow 50$

* Find min and Max

(i) min = 8

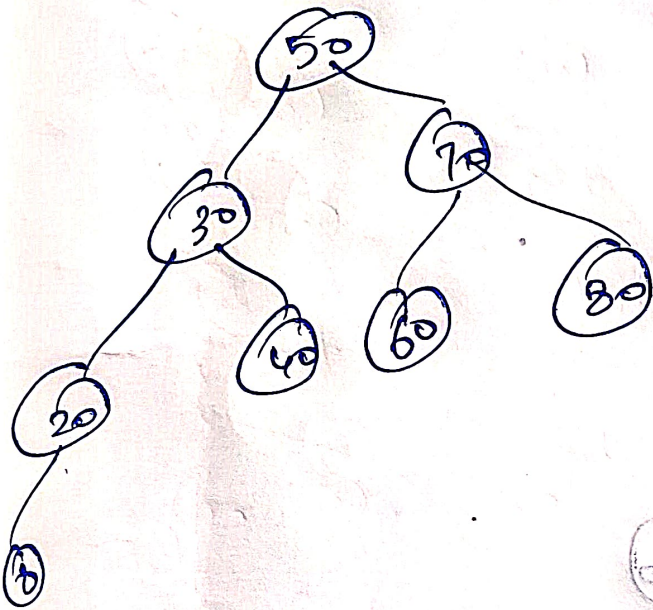
(ii) max = 80

* Delete the node and explain the case used.

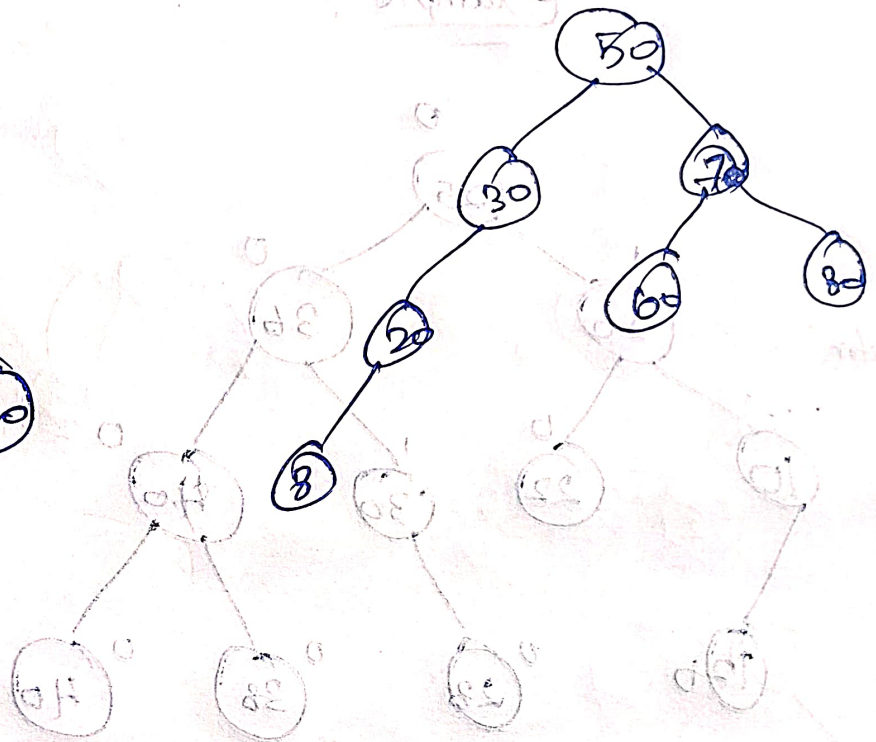
Delete 40

→ 40 is a leaf node, so this is Case 1 in deletion of a leaf node.

Before Deletion



After Deletion



- > -

Rotation

Rotation is the process of moving nodes

around root to make the tree balanced

(Left-Right)

Right-Left

(Right-Right)

Left-Left